

InterVA5: SCI (probbase) & Data Checks

South Africa openVA Training workshop

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InterVA5: Probbase

- ▶ probbase – Symptom Cause Information mapping symptoms to causes
 - ▶ MS Excel spreadsheet distributed with Peter Byass' software
 - ▶ also available in InterVA5 R package

```
library(InterVA5)
data(probbaseV5)
dim(probbaseV5)
```

```
## [1] 354 87
```

```
colnames(probbaseV5)
```

```
## [1] "indic"      "qdesc"      "sdesc"      "who_2016"   "ilab"       "subst"
## [7] "samb"       "dontask1"   "dontask2"   "dontask3"   "dontask4"   "dontask5"
## [13] "dontask6"   "dontask7"   "dontask8"   "doaskif"    "nnonly"     "a_nrp"
## [19] "a_pend_6w"  "a_preg"     "b_0101"     "b_0102"     "b_0103"     "b_0104"
## [25] "b_0105"     "b_0106"     "b_0107"     "b_0108"     "b_0109"     "b_0110"
## [31] "b_0111"     "b_0112"     "b_0199"     "b_0201"     "b_0202"     "b_0203"
## [37] "b_0204"     "b_0205"     "b_0299"     "b_0301"     "b_0302"     "b_0303"
## [43] "b_0401"     "b_0402"     "b_0403"     "b_0499"     "b_0501"     "b_0502"
## [49] "b_0601"     "b_0602"     "b_0701"     "b_0801"     "b_0901"     "b_0902"
## [55] "b_0903"     "b_0904"     "b_0905"     "b_0906"     "b_0907"     "b_0908"
## [61] "b_0999"     "b_1001"     "b_1002"     "b_1003"     "b_1004"     "b_1006"
## [67] "b_1099"     "b_1101"     "b_1102"     "b_1201"     "b_1202"     "b_1203"
## [73] "b_1204"     "b_1205"     "b_1206"     "b_1207"     "b_1208"     "b_1209"
## [79] "b_1210"     "b_1299"     "b_9800"     "c_cult"     "c_emer"     "c_hsys"
## [85] "c_inev"     "c_know"     "c_resr"
```

InterVA5: Probbase (cont.)

- ▶ Basic layout
 - ▶ rows for each symptom (s)
 - ▶ causes of death (c)
 - ▶ cells: $\text{Prob}(s | c)$ = probability of observing s if c is the cause
- ▶ Additional columns for...
 - ▶ names and descriptions of symptoms (InterVA5 & WHO)
 - ▶ data checks (do ask, don't ask, neonate only)
 - ▶ **circumstances of mortality** categories
- ▶ Probabilities in probbases are letter grades, which are converted to probabilities
 - ▶ I = always; $A+$ = almost always; A = 0.5 common;
 $B+$ = often; $C+$ = unusual; $D+$ = very rare;
 E = hardly ever; N = never

Probbase: Grades $\rightarrow$ Probabilities

Grades	Probs	Grades	Probs	Grades	Probs
I	1.0	B	0.050	D+	0.00100
A+	0.8	B-	0.020	D	0.00050
A	0.5	C+	0.010	D-	0.00010
A-	0.2	C	0.005	E	0.00001
B+	0.1	C-	0.002	N	0.00000

Data Checks

- ▶ InSilicoVA & InterVA perform *data consistency checks* to ensure the symptoms do not suggest conflicting information (2 passes through the data). For example:
 - ▶ *Age In Days* 14
 - ▶ *How long did (s)he have a cough?* 4 weeks
- ▶ The software will change these data, but only on a copy of the data (not your data frame that you pass as an argument to codeVA)
- ▶ These changes are described in the log files `errorlogV5.txt` and `errorlog_insilico.txt` (for InterVA5 and InSilicoVA) with the argument

```
results1 <- codeVA(data = data1, data.type = "WH02016",  
                  model = "InSilicoVA",  
                  warning.write = TRUE)
```

- ▶ Info in Error log: (1) record ID; (2) index symptom; and (3) don't ask / ask if / neonate symptom

Data Checks: Three types of consistency checks

1. **Don't ask:** if the index symptom is inconsistent with a “don't ask” symptom, then the index symptom is set to missing
 - ▶ *Error log:* ID – index symptom – don't ask symptom – “cleared in working information”
 - ▶ *inconsistent* index symptom == *substantive* value (as shown in the subst column of probbase.xls – the InterVA5 SCI) and “don't ask” symptom == last character in dontask column
 - ▶ both symptoms need to have non-missing values

Data Checks: Example of don't ask

- ▶ 4 W610059-o (married) value inconsistent with W610022-a (65+) - cleared in working information
 - ▶ record ID is 4
 - ▶ i059o Was she married at the time of death? (with subst == Y)
 - ▶ i022a Was s(he) aged 65 years or more at death?
 - ▶ **don't ask** i059o if i022a == Y
 - ▶ **action:** i059o is changed from Y to missing

Data Checks: Three types of consistency checks

2. **Ask if:** if the index symptom == *substantive* value and the “ask if” symptom != last character in askif column, then assign “ask if” symptom to the last character in askif column
 - ▶ *Error log:* ID – index symptom – ask if symptom – “updated in working information”
 - ▶ index symptom need to have non-missing value

Data Checks: Example of ask if

- ▶ 7 W610152-o (fev nsw) not flagged in category W610147-o (fever) - updated in working information
 - ▶ record ID is 7
 - ▶ i152o Did (s)he have night sweats? (with subst Y)
 - ▶ i147o During the illness that led to death, did (s)he have a fever?
 - ▶ **ask** i152o if i147o == Y
 - ▶ **action:** i147o is changed from missing to Y

Data Checks: Three types of consistency checks

3. **Neonate Only:** if the index symptom == *substantive* value and the deceased was NOT a neonate, then the index symptom is set to missing
- ▶ *Error log:* ID – index symptom – don't ask symptom – "only required for neonates - cleared in working information"
 - ▶ index symptom need to have non-missing value

Data Checks: Example of neonate only

- ▶ 103075 W610394-a (born 1st pr) only required for neonates - cleared in working information
 - ▶ record ID is 103075
 - ▶ i394a Was this baby born from the mother's first pregnancy?
 - ▶ but not a neonate death
 - ▶ **action:** i394a is set to missing

Data Checks: Three types of consistency checks

- ▶ data consistency checks performed in this order
 1. don't ask
 2. ask if
 3. neonates only
- ▶ (then this is repeated with a second pass through the data)

Data Checks: More Examples

- ▶ 60 W610022-f (1 to 11m) value inconsistent with W610022-g (neonate) - cleared in working information
 - ▶ record ID is 60
 - ▶ i022f Was s(he) aged 1 to 11 months at death?
 - ▶ i022g Was s(he) aged < 1 month (28 days) at death?
 - ▶ problem: both symptoms == Y
 - ▶ **action:** i022f is set to missing

Data Checks: More Examples

- ▶ 4 W610304-o (obs pain) value inconsistent with W610022-a (65+) - cleared in working information
 - ▶ do not ask i304o (b/c i022a == Y) so set i304o to missing
- ▶ 4 W610342-o (del norm) value inconsistent with W610022-a (65+) - cleared in working information
 - ▶ do not ask i342o (b/c i022a == Y) so set i342o to missing
- ▶ 7 W610304-o (obs pain) not flagged in category W610297-o (men betw) - updated in working information
 - ▶ ask i304o if i297o == Y
 - ▶ i304o has a value of Y, but i297o is missing, so update i297o to Y
- ▶ 7 W610342-o (del norm) not flagged in category W610313-o (d aft del) - updated in working information
 - ▶ ask i342o if i313o == Y
 - ▶ i342o has a value of Y, but i313o is missing, so update i313o to Y